TECHNICAL PROFESSIONAL

IA - DOMESTIC REFRIGERATION AND AIR-CONDITIONING SERVICING

Grade 12

Three-Term Budget Of Work (BOW) for Learning Competencies

TECHNICAL PROFESSIONAL
IA - INDUSTRIAL TECHNOLOGIES
Grade 12

Three-Term Budget Of Work
For Learning Competencies

Please note that each TechPro elective is completed within one term if delivered at Grade 12.

Term Plan	
Elective	**Domestic Refrigeration and Air-Conditioning Servicing**
Performance Standard	The learner performs the installation of domestic refrigeration and air-conditioning units with safety precautions. The learner performs the maintenance of domestic refrigeration and window-type air- conditioning units with safety precautions. The learner performs troubleshooting of domestic refrigeration and window-type air- conditioning units with safety precautions. The learner performs repairing domestic refrigeration and window-type air-conditioning units with safety precautions.
Content Standard	• Understands the concepts and principles of installing domestic refrigeration and air-conditioning units. • Understands the concepts and principles of maintaining domestic refrigeration and window-type air-conditioning units. • Understands the concepts and principles of troubleshooting domestic refrigeration and window-type air-conditioning units. • Understands the concepts and principles of repairing domestic refrigeration and window-type air-conditioning units.

Week	Learning Competency	Suggested Activities
1	• Discuss overview of Refrigeration and Air Conditioning servicing. • Discuss refrigeration and air conditioning concepts and principles.	• **Service Cycle Exploration** ○ Learners study the basic steps and sequence of refrigeration and air conditioning servicing, including maintenance and troubleshooting tasks. ○ Observe a demonstration or video of the service cycle, then identify and list the key steps involved. • **Servicing Tools Identification** ○ Learners identify common tools and equipment used in refrigeration and air conditioning servicing. ○ Examine each tool, describe its purpose, and demonstrate correct handling. • **Principle Mapping** ○ Learners explore key concepts like the refrigeration cycle, heat transfer, and thermodynamic principles used in air conditioning systems. ○ Create a concept map showing components, flow of refrigerant, and energy transfer in a refrigeration system. • **Case Study Analysis**

Week	Learning Competency	Suggested Activities
		○ Learners analyze real-world air conditioning problems to understand practical applications of refrigeration principles. ○ Review a case study of a malfunctioning AC system and explain which principles are involved in diagnosing the problem.
2 to 3	• Perform pre-installation procedure for an air-conditioning unit.	• **Site Inspection Drill** ○ Learners assess the installation site for window or split-type AC units, checking measurements, power supply, and support structures. ○ Inspect the designated site, measure openings, and verify electrical and structural requirements before installation. • **Component Preparation** ○ Learners prepare AC unit components (filters, brackets, screws, electrical wires) for installation. ○ Collect, organize, and inspect all AC components to ensure readiness for installation.
4 to 5	• Perform procedures in installing a window-type	• **Mounting Practice**

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	air-conditioning unit.	○ Learners practice mounting a window-type AC unit securely in a frame or window opening. ○ Position the AC unit in the window opening, level it, and secure it using brackets and screws. • **Electrical and Functional Check** ○ Learners connect the AC unit to a power supply and verify correct operation, cooling, and safety features. ○ Plug in the unit, turn it on, and check for proper cooling, airflow, and safe electrical function.
6 to 7	• Perform maintenance of window-type air-conditioning units.	• **Air Filter and Coil Cleaning** ○ Learners clean and inspect the air filters and evaporator/condenser coils to ensure proper airflow, efficiency, and unit longevity. ○ Remove the filters and coils, clean them using water, brushes, or compressed air, inspect for damage, and reinstall correctly. • **Electrical and Drainage System Check** ○ Learners inspect the unit's electrical

Week	Learning Competency	Suggested Activities
		connections, controls, and drainage system to ensure safe operation and proper water flow. ○ Check wiring and terminals, test fan and thermostat functions, and clear the drain pan and pipe of debris.
8 to 9	• Discuss pre-troubleshooting of domestic refrigeration and window-type air conditioning units. • Perform troubleshooting of domestic refrigeration and window-type air-conditioning units.	• **Common Fault Identification** ○ Learners study typical problems in domestic refrigerators and window-type AC units, such as poor cooling, strange noises, leaks, and electrical issues. ○ Examine demonstration units or case examples and list common faults, their possible causes, and observable symptoms. • **Troubleshooting Checklist Creation** ○ Learners develop a pre-troubleshooting checklist to guide systematic inspection before performing maintenance or repairs. ○ Create a step-by-step checklist covering electrical, mechanical, and operational aspects of the units to identify potential issues efficiently.

Week	Learning Competency	Suggested Activities

		• **Fault Diagnosis Drill**
		  ○ Learners identify and diagnose common operational problems in domestic refrigerators and window-type AC units, such as cooling issues, leaks, or abnormal noises.
		  ○ Inspect the unit, test electrical and mechanical components, and determine the root cause of the malfunction.
		• **Problem-Solving Simulation**
		  ○ Learners simulate troubleshooting scenarios to practice correcting faults safely, including replacing or repairing defective components.
		  ○ Follow a systematic troubleshooting sequence using the checklist, perform necessary repairs or adjustments, and verify the unit operates correctly.

10 to 11	• Discuss repairing of domestic refrigeration and window-type air conditioning units. • Perform repairing of domestic refrigeration and window-type air-conditioning units.	• **Component Repair Analysis** ○ Learners study the repair procedures for common components of domestic refrigerators and window-type AC units, such as compressors, thermostats, fans, and coils.

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		○ Examine sample components, identify possible faults, and explain step-by-step repair methods and safety precautions. • **Repair Procedure Demonstration** ○ Learners observe and discuss actual repair techniques for refrigeration and AC units, including replacing faulty parts and restoring proper operation. ○ Watch a guided demonstration of repairs, take notes on procedures and tools used, and explain the sequence of steps required for safe and effective repair. • **Component Replacement Practice** ○ Learners practice repairing defective components such as thermostats, compressors, fans, and electrical connections in refrigeration and window-type AC units. ○ Identify the faulty component, safely remove it, replace it with a new or repaired part, and reassemble the unit correctly. • **Functional Restoration Drill**

Week	Learning Competency	Suggested Activities
		○ Learners perform end-to-end repair procedures, ensuring the refrigeration or AC unit is restored to proper operation and tested for safety and efficiency. ○ Follow a step-by-step repair sequence, test all functions, and verify that cooling, airflow, and electrical systems operate correctly.
Suggested Performance Tasks	• **Individual Performance Task - Window-Type AC Installation** ○ Each learner independently installs a window-type air-conditioning unit, ensuring correct placement, secure mounting, proper electrical connection, and adherence to safety procedures. ○ Measure and prepare the installation site, mount the unit securely, connect the power supply safely, and test for proper operation. • **Group Performance Task - Domestic Refrigerator Setup Project** ○ In groups, learners collaboratively install a domestic refrigeration unit, dividing tasks such as positioning, leveling, electrical connection, and safety checks, ensuring proper installation and operation. ○ Work as a team to position the refrigerator, connect power safely, verify leveling and clearance, and test all functions according to safety standards. • **Individual Performance Task - Routine AC and Refrigerator Maintenance** ○ Each learner independently performs routine maintenance on a domestic refrigerator or window-type AC unit, including cleaning filters, coils, and drainage, and checking electrical connections safely. ○ Follow a step-by-step maintenance procedure, clean and inspect all components, and ensure proper operation while observing safety precautions.	

Week	Learning Competency	Suggested Activities
		• **Group Performance Task - Comprehensive Maintenance Workshop** ○ In groups, learners collaboratively perform full maintenance on multiple refrigeration and AC units, dividing tasks such as coil cleaning, filter replacement, electrical inspection, and drainage verification, while following safety protocols. ○ Work as a team to perform thorough maintenance on assigned units, checking all mechanical and electrical systems, and confirm proper operation safely. • **Individual Performance Task - Fault Diagnosis Challenge** ○ Each learner independently troubleshoots a malfunctioning domestic refrigerator or window-type AC unit, identifying electrical, mechanical, or operational faults while following safety protocols. ○ Inspect the unit systematically, test components, identify the root cause of the problem, and record findings safely. • **Group Performance Task - Collaborative Troubleshooting Exercise** ○ In groups, learners work together to troubleshoot multiple refrigeration and AC units, sharing responsibilities such as checking circuits, testing compressors, and verifying airflow, ensuring safe procedures throughout. ○ Divide tasks among team members to diagnose all system components, identify faults, and recommend corrective actions while observing all safety precautions. • **Individual Performance Task - Component Repair Challenge** ○ Each learner independently repairs a defective component (e.g., thermostat, fan, compressor, or electrical connection) in a domestic refrigerator or window-type AC unit while following safety protocols. ○ Identify the faulty component, perform the repair or replacement safely, and test the unit to ensure proper operation. • **Group Performance Task - Full-System Repair Project** ○ In groups, learners collaboratively repair multiple refrigeration or AC units, dividing tasks for diagnosing faults, replacing defective parts, and restoring full functionality safely.

Week	Learning Competency	Suggested Activities
		○ Work as a team to troubleshoot, repair, and test each unit, ensuring all electrical and mechanical systems are safely restored to proper operation.